Faculty of Computer & Information Sciences
Ain Shams University
Course Code: SCO 421
Course Name: Computer Vision
Exam: (MidTerm) 14/11/2023
Year: (1^{st} term) 4^{th} undergraduate (SC-CSys)

Instructors: Dr. Dina Khattab
Offering Dept: Scientific Computing
Academic year: 2023-2024
Duration: 30 Minutes
Total Grade: 15 Marks

الاسم:	....	**المستوي:**	
القسم:	....	**رقم الجلوس:**	....

A. [8 Marks] State whether each of the following statements is True of False.

1. When filtering an image with a Gaussian filter, the filter size is the most important because it defines the smoothing scale. x
2. Averaging is considered an intuitive solution for image smoothing as it assumes the independence of the noise added to the image. √
3. Linear filtering complexity is dependent on both image size and filter size. √
4. Sobel filters are typically used as edge detectors rather than sharpening tools. √
5. Canny edge detector is based on the 1^{st} derivative of a Gaussian. √
6. It is important to convolve an image with a Gaussian filter before convolving with a Laplacian filter. x
7. For finding horizontal edges in an image, we use the following filter $\begin{matrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{matrix}$ x
8. In canny edge detector, reducing both the high and low thresholds of the threshold hysteresis step, will get more discontinuous edges. x

B. [3 Marks] Using the second finite difference operator

-1	2	-1

for edge detection, show how the pixel values in the top row of the following 1D signal are changed by discrete convolution with this operator (insert the output values in the bottom row). State if alternatively applying correlation with the operator would change the computed results? Why?

input	0	0	0	0	5	5	5	5	5	0	0	0	0
output	0	0	0	-5	5	0	0	0	5	-5	0	0	0

- No
- Because filter is symmetric

C. [4 Marks] Given the following pairs of images, we need to apply image matching between images of each pair. The first step is to apply feature detection and use these features for the matching process.

i. Which pair do you think that Harris detector will be appropriate for this mission? Explain why?

Pair 1

Pair 2

ii. Given the following Moment matrix M by Harris detector, do you think that it corresponds to a corner? Explain why?

$$M = \begin{bmatrix} 29 & 97 \\ 97 & 353 \end{bmatrix}$$

i. Pair 1

Because Harris is invariant to rotation but not to scale.

ii. No

because Iy >> Ix

With Best Regards,

Dr. Dina Khattab